

Администрирование локальных сетей

Настройка сетевых сервисов. DHCP

Пакавира Арсениу Висенте Луиш

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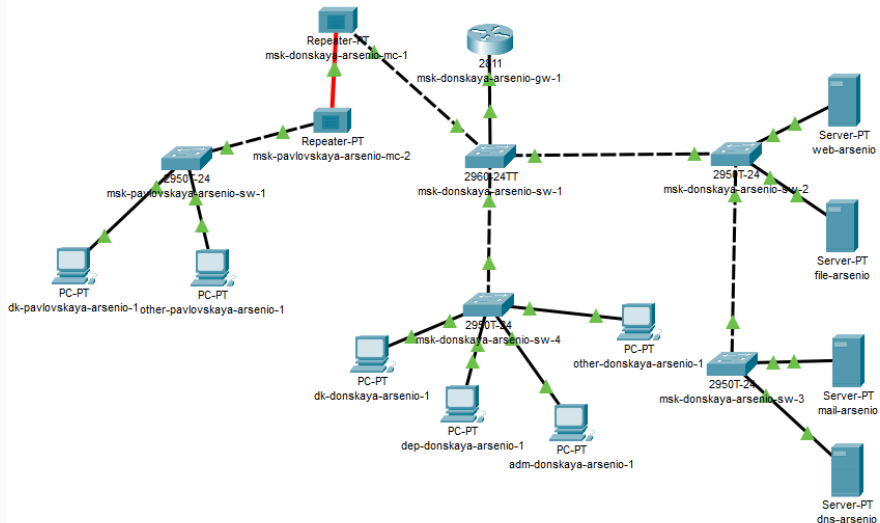
Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Приобрести практические навыки настройки динамического распределения IP-адресов посредством протокола DHCP в локальной сети.

Настройка DNS-сервера

Добавление DNS-сервера



Настройка IP-адреса DNS-сервера

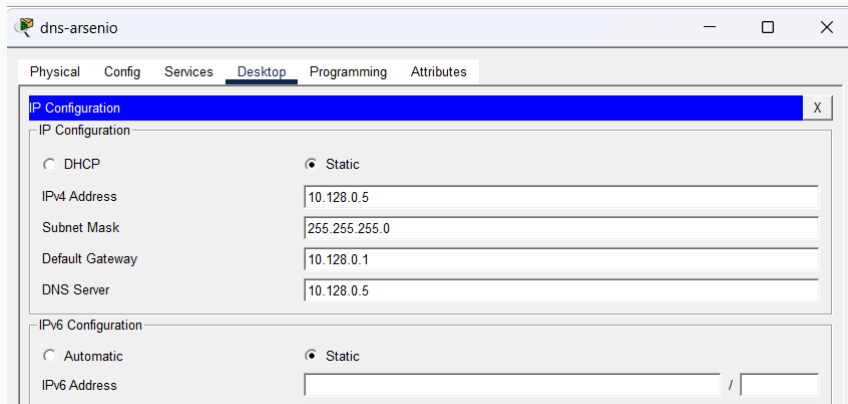


Рис. 2: Настройка статического IP-адреса на DNS-сервере

Настройка DNS-записей

The screenshot shows the 'dns-arsenio' web interface. The 'Services' tab is selected, and the 'DNS' service is enabled. The 'Resource Records' section shows a table with four A records. The 'Add' button is highlighted.

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type

Address

No.	Name	Type	Detail
0	dns.donskaya.rudn.ru	A Record	10.128.0.5
1	file.donskaya.rudn.ru	A Record	10.128.0.3
2	mail.donskaya.rudn.ru	A Record	10.128.0.4
3	www.donskaya.rudn.ru	A Record	10.128.0.2

Рис. 3: Настройка DNS-записей на сервере dns-arsenio

Настройка DHCP

DHCP-пулы на маршрутизаторе

```
Password:
msk-donskaya-arsenio-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-arsenio-gw-1(config)#
msk-donskaya-arsenio-gw-1(config)#ip name-serv 10.128.0.5
msk-donskaya-arsenio-gw-1(config)#service dhcp
msk-donskaya-arsenio-gw-1(config)#ip dhcp pool dk
msk-donskaya-arsenio-gw-1(dhcp-config)#net 10.128.3.0 255.255.255.0
msk-donskaya-arsenio-gw-1(dhcp-config)#default-router 10.128.3.1
msk-donskaya-arsenio-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-arsenio-gw-1(dhcp-config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-arsenio-gw-1(config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-arsenio-gw-1(config)#
msk-donskaya-arsenio-gw-1(config)#ip dhcp pool departaments
msk-donskaya-arsenio-gw-1(dhcp-config)#net 10.128.4.0 255.255.255.0
msk-donskaya-arsenio-gw-1(dhcp-config)#default-router 10.128.4.1
msk-donskaya-arsenio-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-arsenio-gw-1(dhcp-config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-arsenio-gw-1(config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-arsenio-gw-1(config)#
msk-donskaya-arsenio-gw-1(config)#ip dhcp pool adm
msk-donskaya-arsenio-gw-1(dhcp-config)#net 10.128.5.0 255.255.255.0
msk-donskaya-arsenio-gw-1(dhcp-config)#default-router 10.128.5.1
msk-donskaya-arsenio-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-arsenio-gw-1(dhcp-config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-arsenio-gw-1(config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
msk-donskaya-arsenio-gw-1(config)#
msk-donskaya-arsenio-gw-1(config)#ip dhcp pool other
msk-donskaya-arsenio-gw-1(dhcp-config)#net 10.128.6.0 255.255.255.0
msk-donskaya-arsenio-gw-1(dhcp-config)#default-router 10.128.6.1
msk-donskaya-arsenio-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-arsenio-gw-1(dhcp-config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-arsenio-gw-1(config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-arsenio-gw-1(config)#
msk-donskaya-arsenio-gw-1(config)#^Z
msk-donskaya-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-gw-1#
```

Получение адреса по DHCP

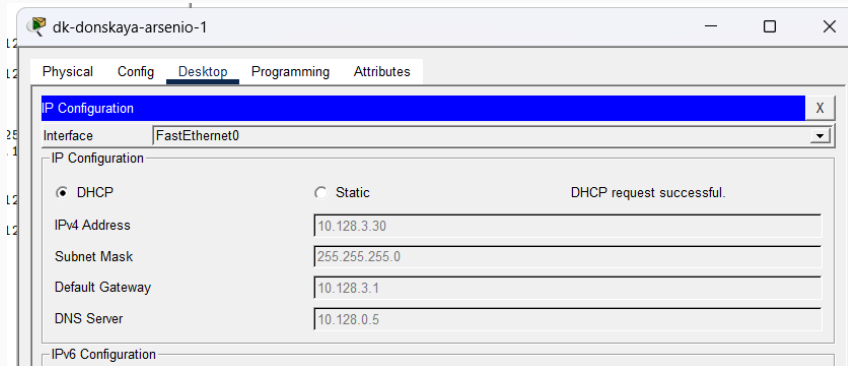


Рис. 5: Получение IP-адреса по DHCP на оконечном устройстве

Проверка доступности подсетей

```
dk-donskaya-arsenio-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.4.30

Pinging 10.128.4.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.3.31

Pinging 10.128.3.31 with 32 bytes of data:

Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128
Reply from 10.128.3.31: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.31:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.5.30

Pinging 10.128.5.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Информация о DHCP-пулах

```
msk-donskaya-arsenio-gw-1#sh ip dhcp pool
Pool dk :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 2
  Excluded addresses : 8
  Pending event : none

  1 subnet is currently in the pool
  Current index IP address range Leased/Excluded/Total
  10.128.3.1 10.128.3.1 - 10.128.3.254 2 / 8 / 254

Pool departments :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Excluded addresses : 8
  Pending event : none

  1 subnet is currently in the pool
  Current index IP address range Leased/Excluded/Total
  10.128.4.1 10.128.4.1 - 10.128.4.254 1 / 8 / 254

Pool adm :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Excluded addresses : 8
  Pending event : none

  1 subnet is currently in the pool
  Current index IP address range Leased/Excluded/Total
  10.128.5.1 10.128.5.1 - 10.128.5.254 1 / 8 / 254

Pool other :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 2
  Excluded addresses : 8
  Pending event : none

  1 subnet is currently in the pool
  Current index IP address range Leased/Excluded/Total
  10.128.6.1 10.128.6.1 - 10.128.6.254 2 / 8 / 254
```

```
msh-donskaya-arsenio-gw-1#  
msh-donskaya-arsenio-gw-1#sh ip dhcp bin  
IP address      Client-ID/  
                Hardware address      Lease expiration      Type  
10.128.3.30     000D.BD55.BCCC          --                    Automatic  
10.128.3.31     000C.8503.CCD4          --                    Automatic  
10.128.4.30     000B.BE90.DDA4          --                    Automatic  
10.128.5.30     0060.7097.1D63          --                    Automatic  
10.128.6.30     0009.7C76.0418          --                    Automatic  
10.128.6.31     00D0.58C5.357C          --                    Automatic  
msh-donskaya-arsenio-gw-1#
```

Рис. 8: Просмотр привязок выданных DHCP-адресов

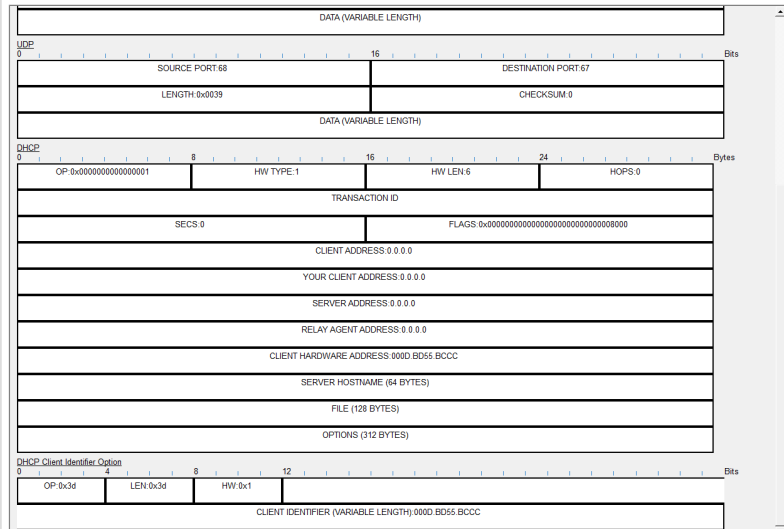
Анализ DHCP в режиме Simulation

DHCP-запрос на маршрутизаторе

PDU Information at Device: msk-donskaya-arsenio-gw-1

OSI Model [Inbound PDU Details](#)

PDU Formats



DHCP-ответ с назначенным адресом

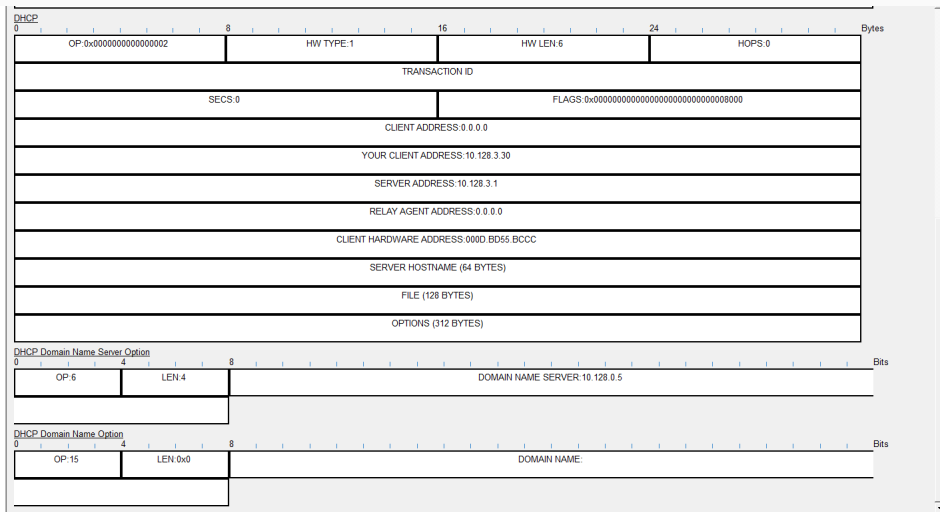


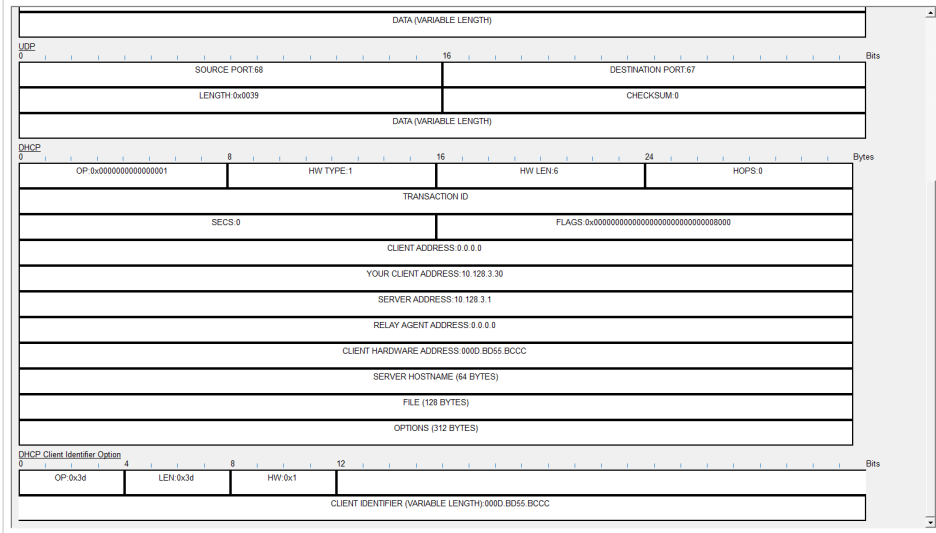
Рис. 10: Анализ DHCP-ответа с назначенным IP-адресом

Передача DHCP через коммутатор

PDU Information at Device: msk-donskaya-arsenio-sw-1

OSI Model [Inbound PDU Details](#) Outbound PDU Details

PDU Formats

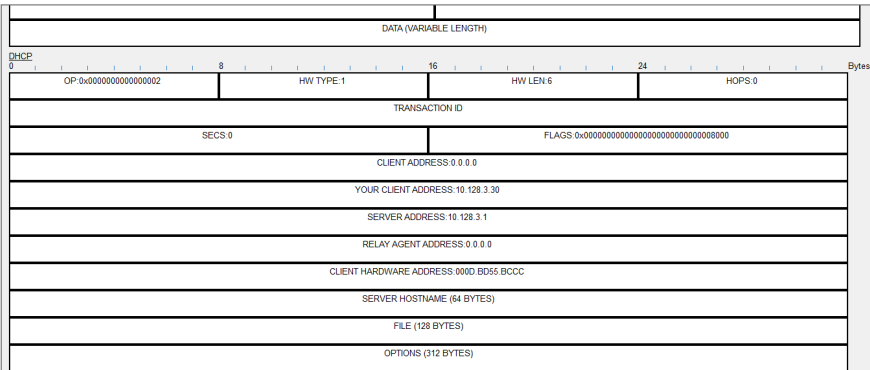


DHCP-ответ на оконечном устройстве

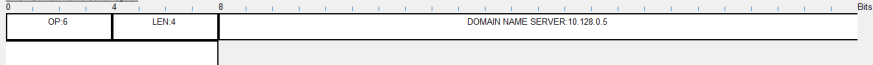
PDU Information at Device: dk-donskaya-arsenio-1

OSI Model [Inbound PDU Details](#)

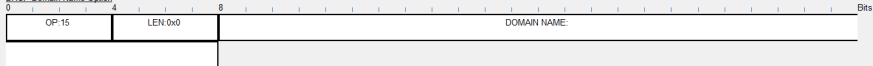
PDU Formats



DHCP Domain Name Server Option



DHCP Domain Name Option



События DHCP в Simulation Panel

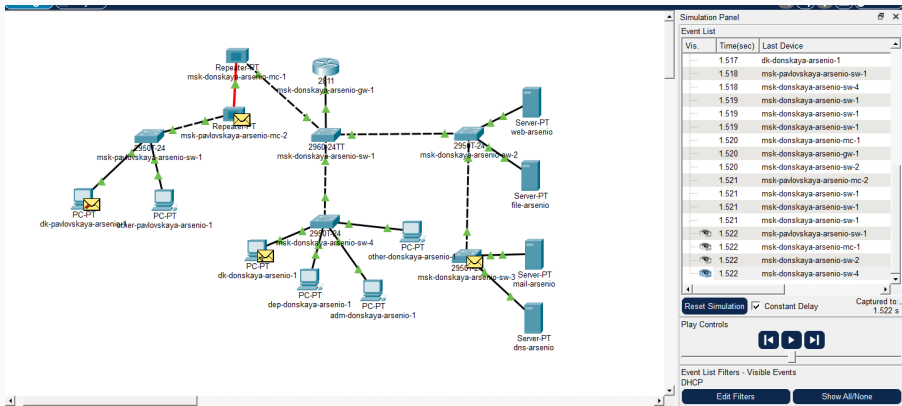


Рис. 13: Просмотр DHCP-событий в режиме Simulation

Выводы

В ходе лабораторной работы был добавлен и настроен DNS-сервер с адресом 10.128.0.5.

На маршрутизаторе были созданы DHCP-пулы для нескольких подсетей, указаны шлюзы, DNS-сервер и исключённые диапазоны адресов.

Оконечные устройства успешно получили IP-адреса автоматически, а проверка связи подтвердила доступность узлов из разных подсетей.

В режиме Simulation был изучен процесс DHCP-обмена между клиентом, коммутаторами и маршрутизатором.